

Timing Accuracy and Jitter Characterization of ESP32-Based Phase-Angle AC Control: MicroPython vs. Native C

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ABSTRACT

This study provides a quantitative comparison between MicroPython and native C (using ESP-IDF) implementations for phase-angle AC control on the ESP32 platform. Measured mean timing errors for MicroPython ranged from 218.2 to 234.7 microseconds, while native C errors were reduced to -10.3 to 6.1 microseconds after calibration. Residual short-term jitter remained similar across both environments (2.5 to 10.1 microseconds standard deviation), as it is primarily dominated by the hardware detection and triggering chain.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Directly justifies the choice of Native C/C++ over MicroPython or high-level scripting for the real-time stepper control loop on the ESP32 S3 supervisor.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Luis E. Bañuelos García provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.